Microeconomics
Course Code: Hum 2101

Lecture 5: Elasticity

Elasticity

Elasticity is a general concept that can be used to quantify the response in one variable when the other variable changes.

The elasticity of A with respect to B $\frac{\% \ \Delta A}{\% \ \Delta B}$

Elasticity is a measure of how much buyers and sellers respond to changes in market conditions.

Elasticity - A measure of the responsiveness of quantity demanded or quantity supplied to a change in one of its determinants

Elasticity

Some purchases, like those for vacation travel, are luxuries that are very sensitive to price changes.

Others, like food or electricity, are necessities for which consumer quantities respond very little to price changes.

The Price Elasticity of Demand

A measure of how much the quantity demanded of a good responds to a change in the price of that good, computed as the percentage change in quantity demanded divided by the percentage change in price.

The **price elasticity of demand** measures how much the quantity demanded responds to a change in price. Demand for a good is said to be *elastic* if the quantity demanded responds substantially to changes in the price. Demand is said to be *inelastic* if the quantity demanded responds only slightly to changes in the price.

Determinants of The Price Elasticity of Demand

- **Availability of Close Substitutes:** Goods with close substitutes tend to have more elastic demand because it is easier for consumers to switch from that good to another. For example, butter and margarine are easily substitutable. A small increase in the price of butter, assuming the price of margarine is held fixed, causes the quantity of butter sold to fall by a large amount. By contrast, because eggs are food without a close substitute, the demand for eggs is less elastic than the demand for butter. A small increase in the price of eggs does not cause a sizable drop in the quantity of eggs sold.
- **Necessities versus Luxuries:** Necessities tend to have inelastic demands, whereas luxuries have elastic demands. When the price of a doctor's visit rises, people will not dramatically reduce the number of times they go to the doctor, although they might go somewhat less often. By contrast, when the price of sailboats rises, the quantity of sailboats demanded falls substantially. The reason is that most people view doctor visits as a necessity and sailboats as a luxury.

Determinants of The Price Elasticity of Demand

- **Definition of the Market:** The elasticity of demand in any market depends on how we draw the boundaries of the market. Narrowly defined markets tend to have more elastic demand than broadly defined markets because it is easier to find close substitutes for narrowly defined goods. For example, food, a broad category, has a fairly inelastic demand because there are no good substitutes for food. Ice cream, a narrower category, has a more elastic demand because it is easy to substitute other desserts for ice cream. Vanilla ice cream, a very narrow category, has a very elastic demand because other flavors of ice cream are almost perfect substitutes for vanilla.
- **Time Horizon:** Goods tend to have more elastic demand over longer time horizons. When the price of gasoline rises, the quantity of gasoline demanded falls only slightly in the first few months. Over time, however, people buy more fuel-efficient cars, switch to public transportation, and move closer to where they work. Within several years, the quantity of gasoline demanded falls more substantially.

Computing the Price Elasticity of Demand

Economists compute the price elasticity of demand as the percentage change in the quantity demanded divided by the percentage change in the price.

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}.$$

For example, suppose that a 10 percent increase in the price of an ice cream cone causes the amount of ice cream you buy to fall by 20 percent. We calculate your elasticity of demand as

$$\text{Price elasticity of demand} = \frac{20 \text{ percent}}{10 \text{ percent}} = 2.$$

The Variety of Demand Curves

Economists classify demand curves according to their elasticity.

Demand is considered elastic when the elasticity is greater than 1, which means the quantity moves proportionately more than the price.

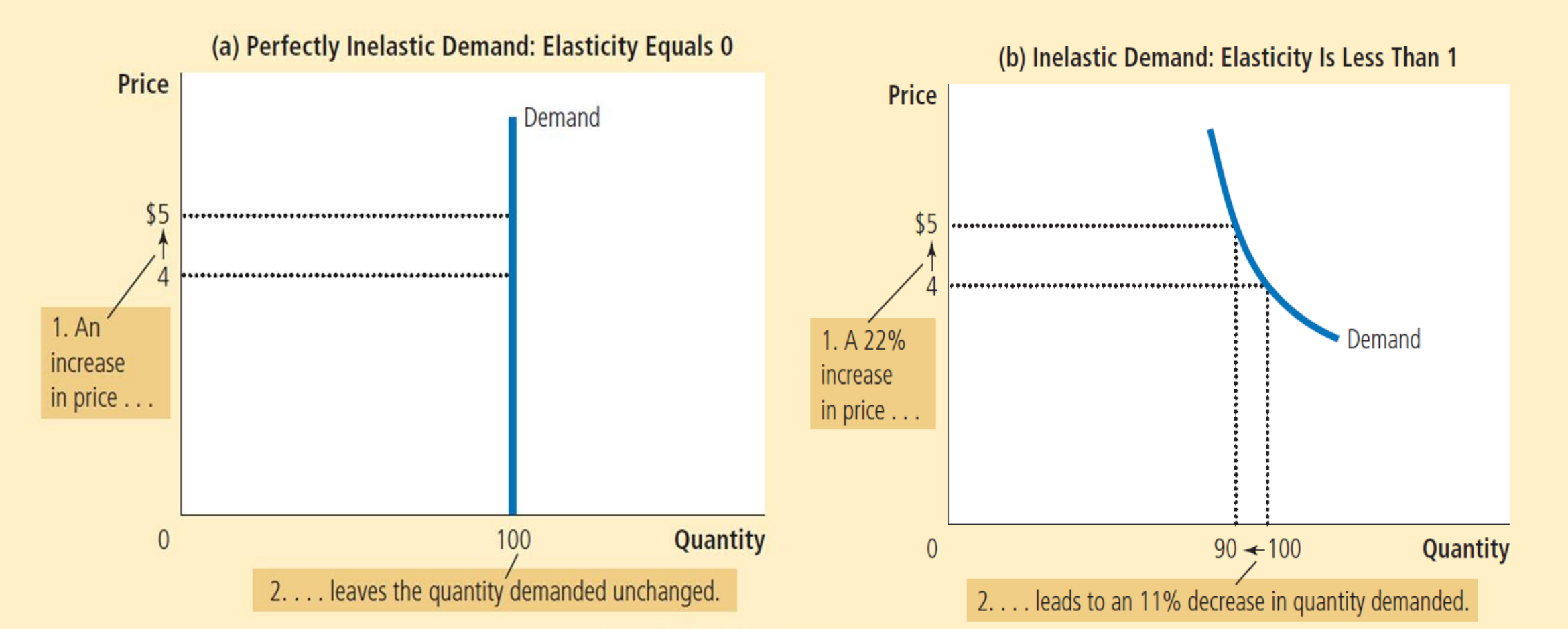
Demand is considered inelastic when the elasticity is less than 1, which means the quantity moves proportionately less than the price.

If the elasticity is exactly 1, the percentage change in quantity equals the percentage change in price, and demand is said to have unit elasticity.

The Variety of Demand Curves

a) Perfectly Inelastic Demand: Elasticity Equals 0
b) Inelastic Demand: Elasticity Is Less Than 1
c) Unit Elastic Demand: Elasticity Equals 1
d) Elastic Demand: Elasticity Is Greater Than 1
e) Perfectly Elastic Demand: Elasticity Equals Infinity

The Variety of Demand Curves



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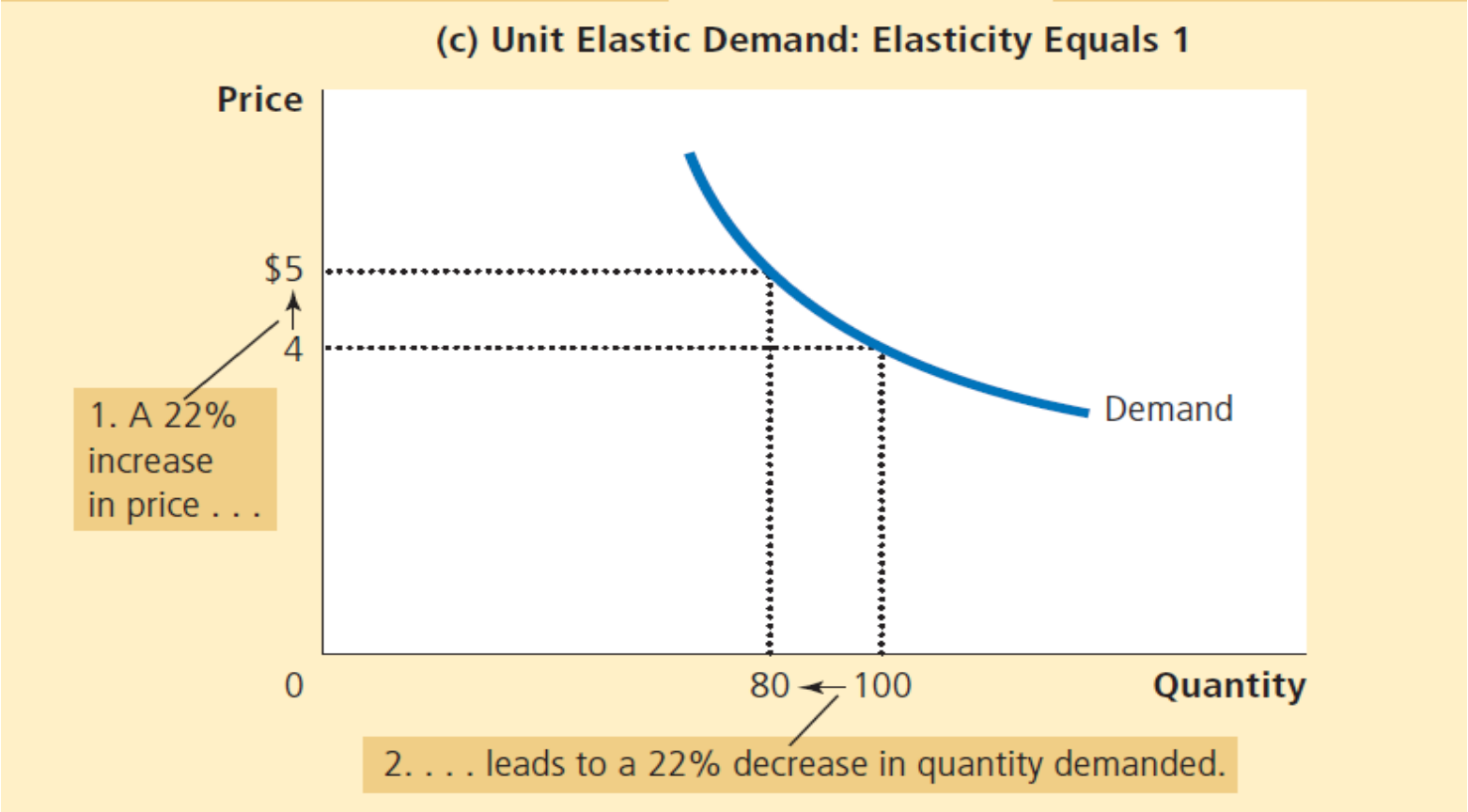

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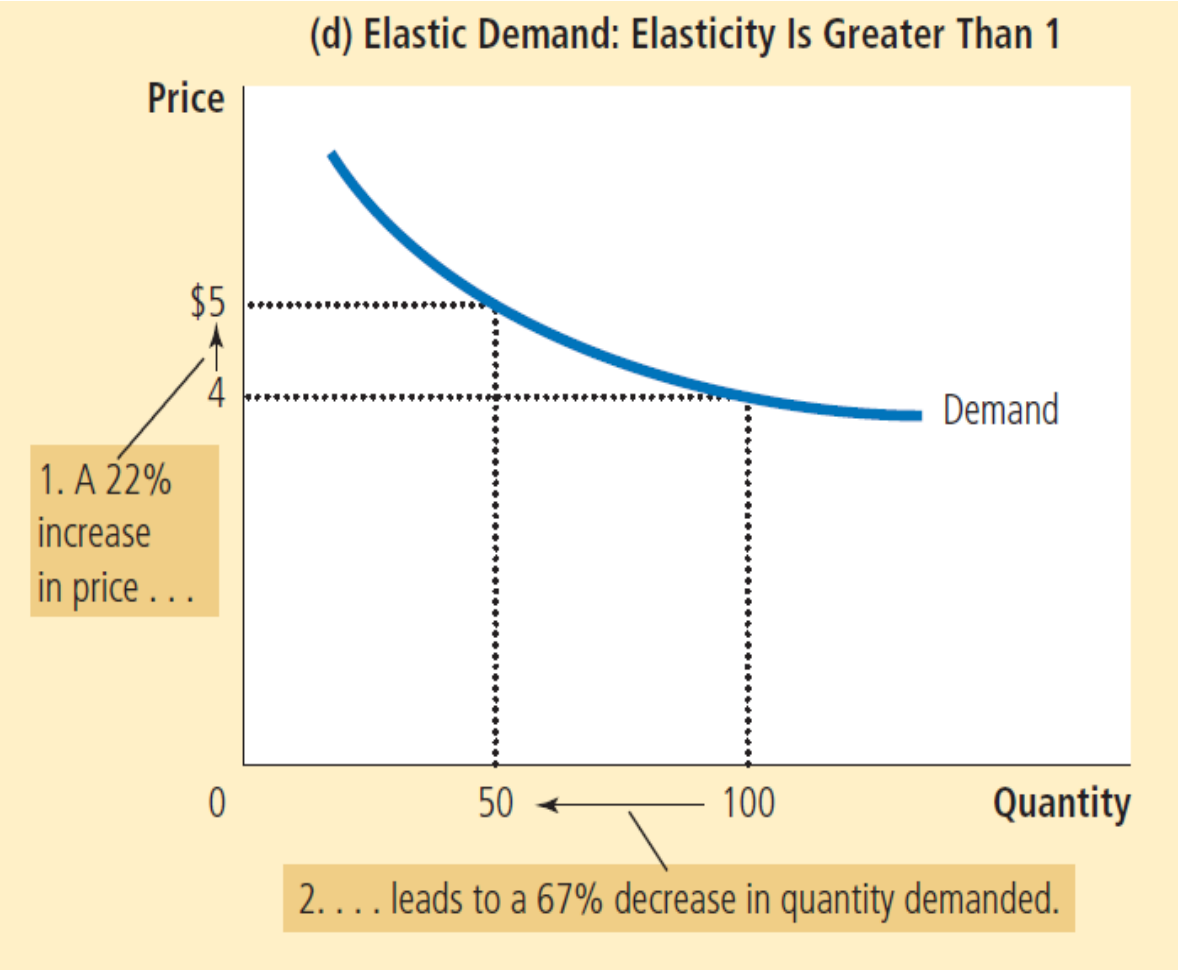

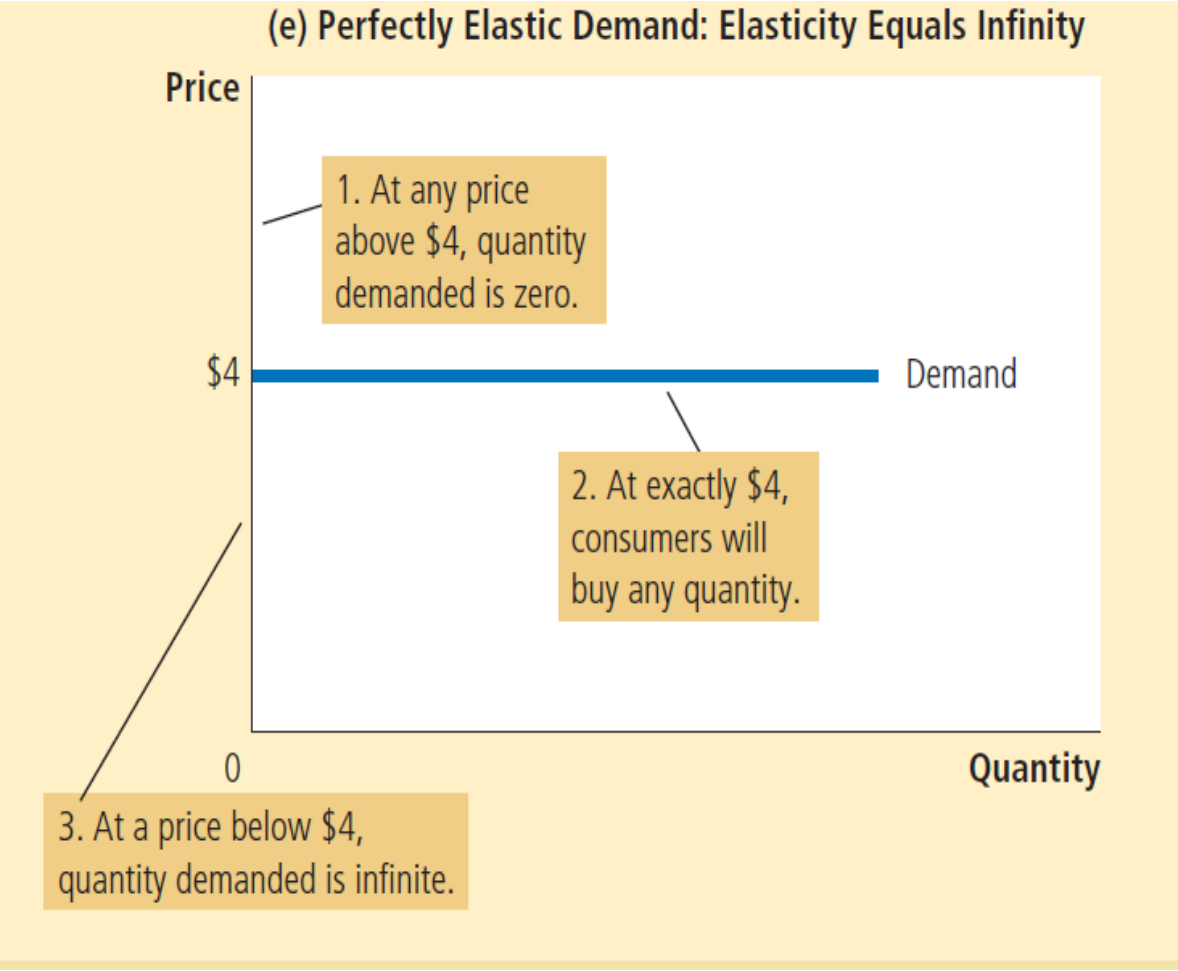

The Price Elasticity of Supply

A measure of how much the quantity supplied of a good responds to a change in the price of that good, computed as the percentage change in quantity supplied divided by the percentage change in price.

The price elasticity of supply measures how much the quantity supplied responds to changes in the price. The supply of a good is said to be elastic if the quantity supplied responds substantially to changes in the price. Supply is said to be inelastic if the quantity supplied responds only slightly to changes in the price.

Determinants of The Price Elasticity of Supply

In most markets, a key determinant of the price elasticity of supply is ***the time period*** being considered. Supply is usually more elastic in the long run than in the short run.

Over short periods of time, firms cannot easily change the size of their factories to make more or less of a good. Thus, in the short run, the quantity supplied is not very responsive to the price.

By contrast, firms can build new factories over longer periods or close old ones. In addition, new firms can enter a market, and old firms can exit. Thus, the quantity supplied can respond substantially to price changes in the long run.

Computing the Price Elasticity of Supply

Economists compute the price elasticity of supply as the percentage change in the quantity supplied divided by the percentage change in the price. That is,

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}.$$

Computing the Price Elasticity of Supply

For example, suppose that an increase in the price of milk from \$2.85 to \$3.15 a gallon raises the amount that dairy farmers produce from 9,000 to 11,000 gallons per month. Using the midpoint method, we calculate the percentage change in price as

$$\text{Percentage change in price} = (3.15 - 2.85)/3.00 \times 100 = 10 \text{ percent.}$$

Similarly, we calculate the percentage change in quantity supplied as

$$\text{Percentage change in quantity supplied} = (11{,}000 - 9{,}000)/10{,}000 \times 100 = 20 \text{ percent.}$$

In this case, the price elasticity of supply is

$$\text{Price elasticity of supply} = \frac{20 \text{ percent}}{10 \text{ percent}} = 2.$$

In this example, the elasticity of 2 indicates that the quantity supplied changes proportionately twice as much as the price.

The Variety of Supply Curves

Economists classify demand curves according to their elasticity.

Supply is considered elastic when the elasticity is greater than 1, which means the quantity moves proportionately more than the price.

Supply is considered inelastic when the elasticity is less than 1, which means the quantity moves proportionately less than the price.

If the elasticity is exactly 1, the percentage change in quantity equals the percentage change in price, and supply is said to have unit elasticity.

The Variety of Supply Curves

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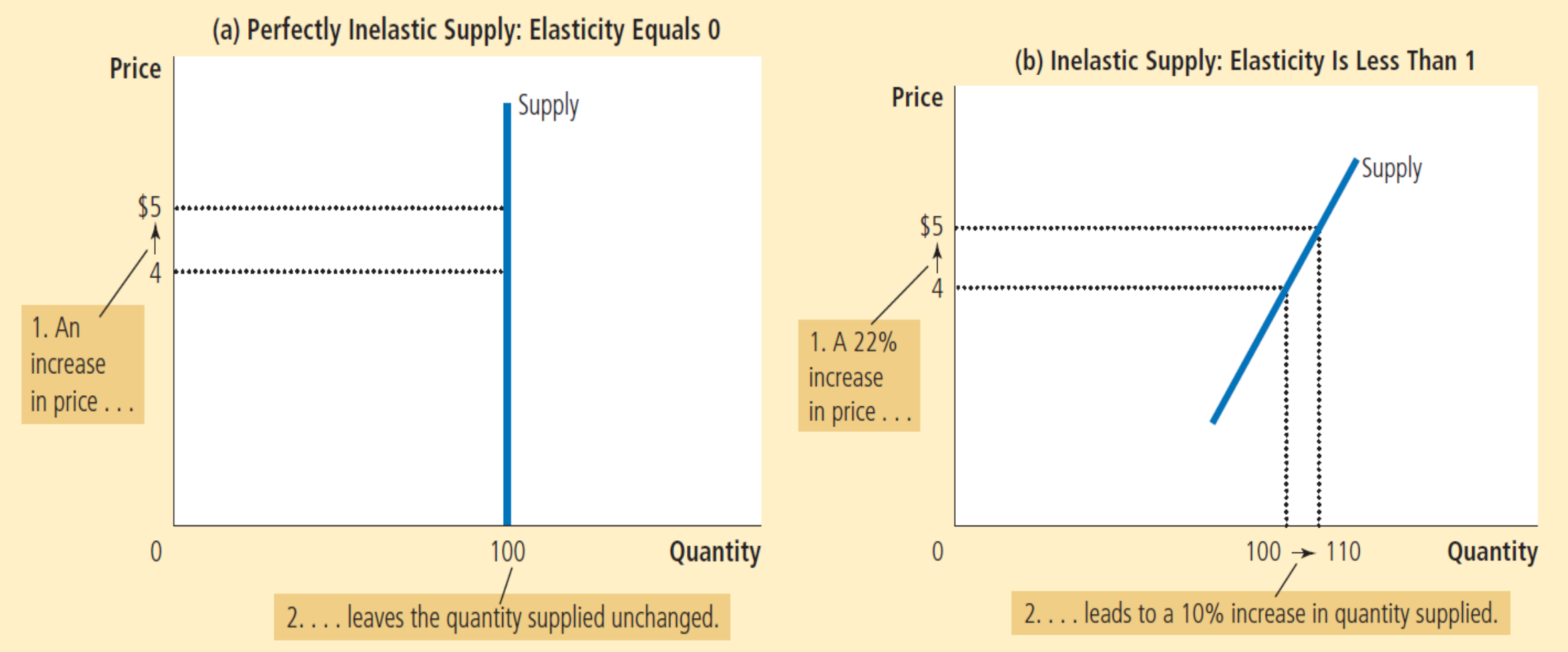
The Variety of Supply Curves



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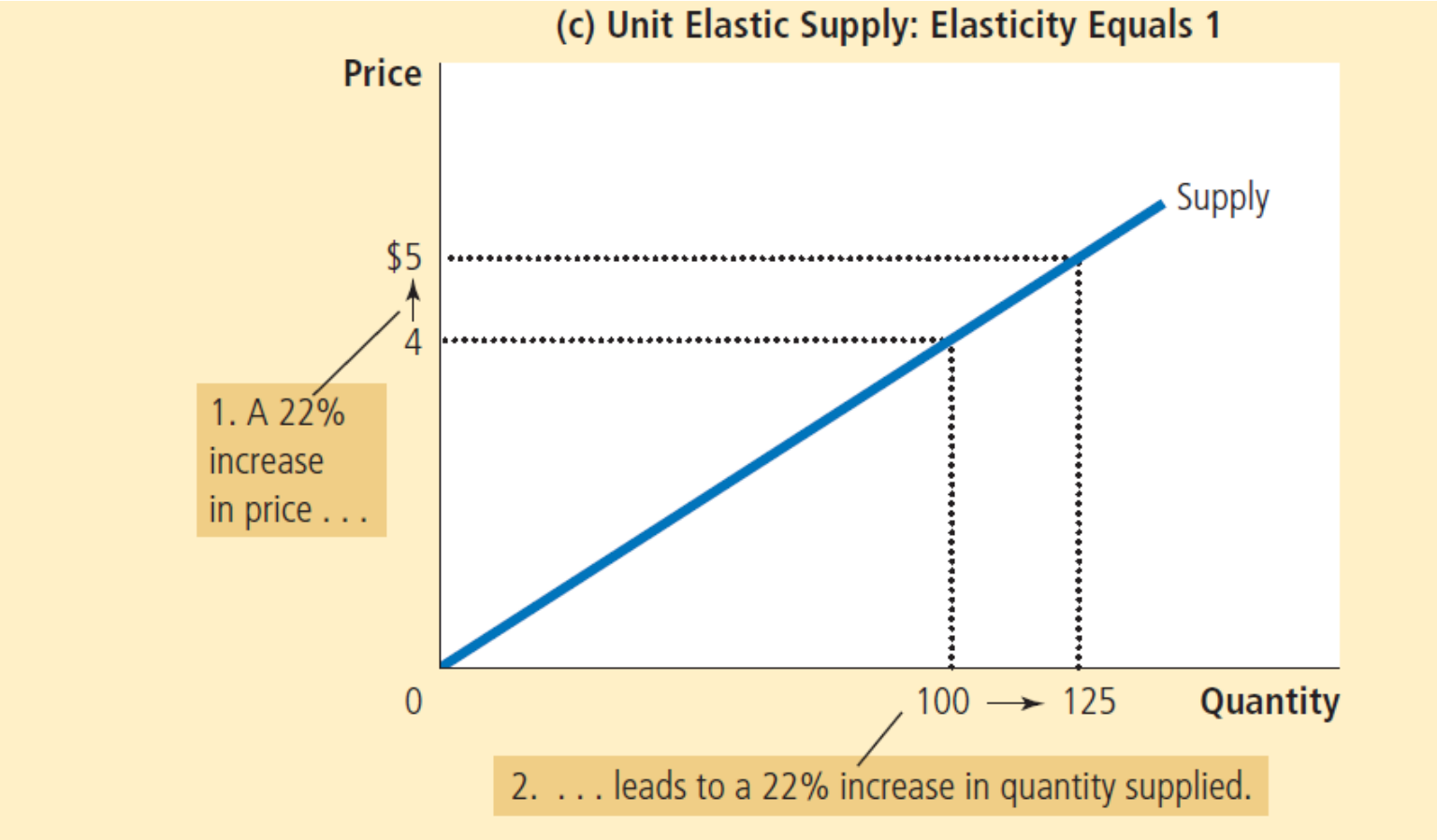
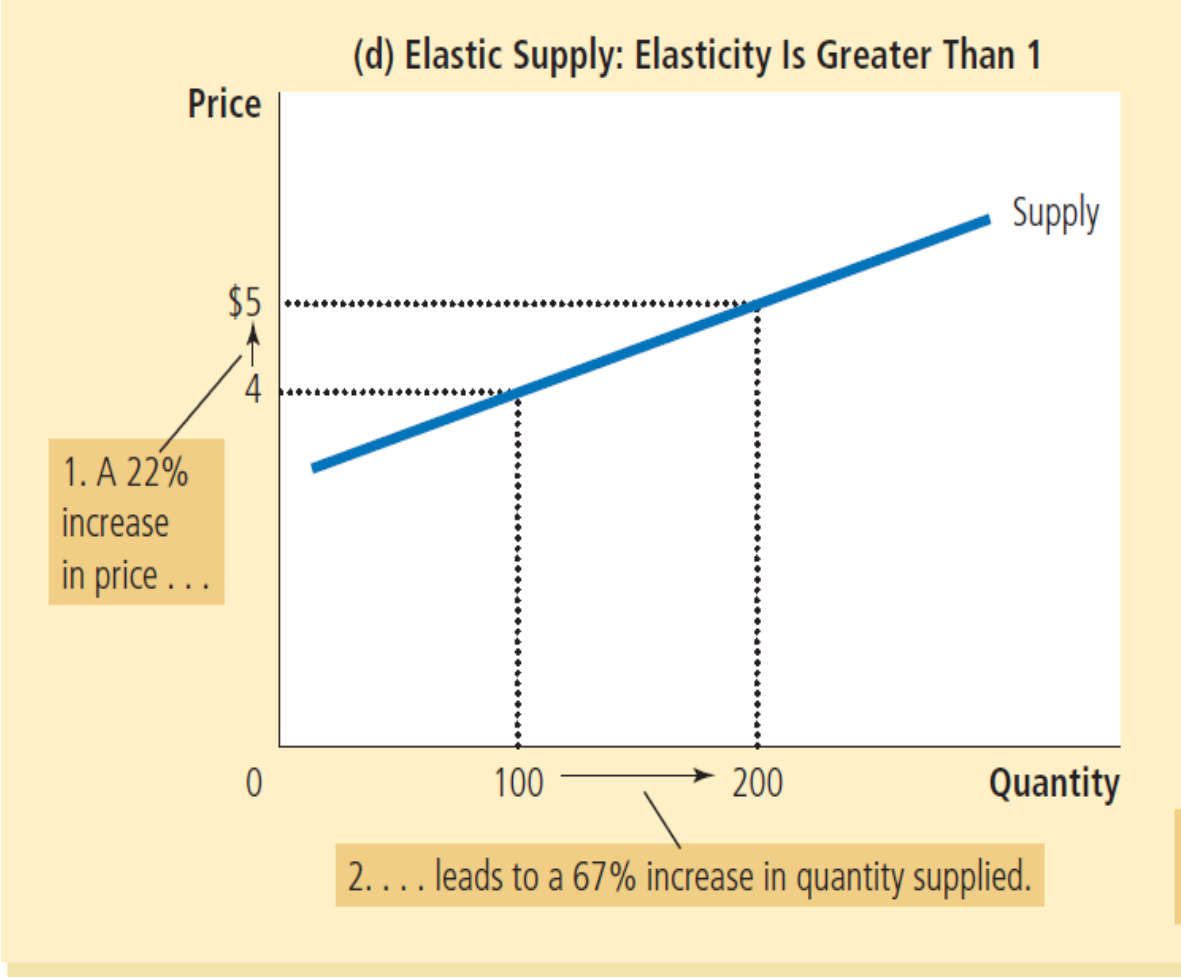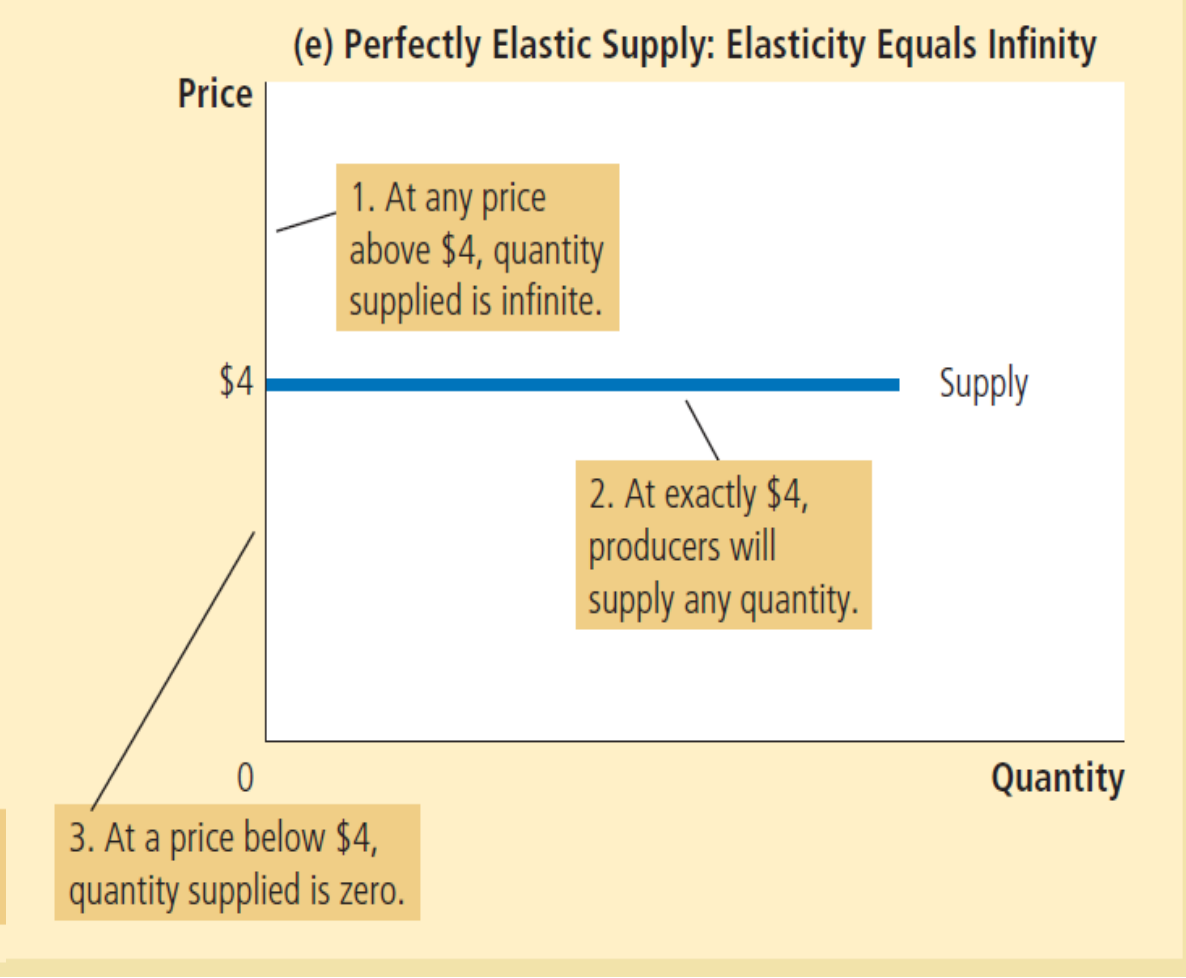
The Variety of Supply Curves